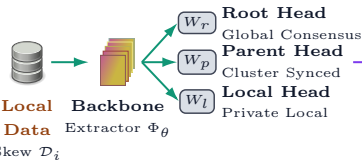


Client-Side Architecture

Single Backbone Φ_θ



ACL M Masking

Zero-gradient shielding on unobserved classes

Skew Head Ensemble

Entropy-weighted inference prediction $\mathbf{z}_{\text{ens}}$

Central Federated Server

Layered Subspace Byzantine Defense and Global Aggregation

Layered Subspace Byzantine Defense

- 1.SCCF**: Subspace Cosine Filtering
 - 2.Norm-Bound**: Adaptive Q_1 Clipping
 - 3.TTT**: Temporal Trust Tracking (τ_{iso})
- Subspace alignment check $\mathcal{S}_i$

Global Aggregation

- 1.Backbone** Φ_θ
- 2.Root Head** W_r

Consensus Broadcast

Trusted fleet $\mathcal{A}^{(t)}$

Privacy Peer Discovery: Spherical k -Means Clustering on Sketches $\{\mathbf{s}_i\}$

Collaborative Peer Clusters



Peer Cluster 1 ($\mathcal{C}_1$)

Parent Head $W_{p,1}$ · Partition 1



Peer Cluster 2 ($\mathcal{C}_2$)

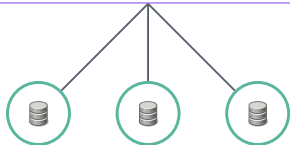
Parent Head $W_{p,2}$ · Partition 2

...



Peer Cluster K ($\mathcal{C}_K$)

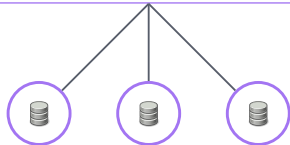
Parent Head $W_{p,K}$ · Partition K



Client 1
Partition 1

Client 2
Partition 1

Client 3
Partition 1

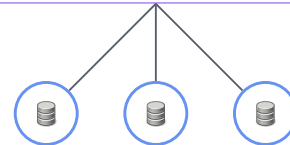


Client 4
Partition 2

Client 5
Partition 2

Client 6
Partition 2

...



Client 7
Partition K

Client 8
Partition K

Client M
Partition K

Legend: → Global Parameter Broadcast -▶ Privacy Sketch Upload → On-Device Feature Forwarding — Peer Cluster Membership